Major Project

Branch: CSE-Data Science

Department: Data Engineering

Semester: VII **Section: A** **Regulation: R23** **Course Code: R23MSCSP003**

Project Title: **AgentIQ:** A AI Framework for Universal File Analysis and Interactive Dashboard Generation over Data

Batch Number: **08**

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Abstract

- AgentIQ is a multi-agent AI framework for data analysis and dashboard generation.
- The system supports different data sources such as CSV, Excel, JSON, Google Sheets, and databases.
- Multiple AI agents work together to understand data and process user queries.
- The framework supports natural language and multilingual query processing.
- It automatically selects suitable analytical methods based on the dataset and user requirements.
- The system generates interactive dashboards and meaningful insights.
- Users can compare, customize, and export different dashboard designs.
- AgentIQ provides an explainable and flexible environment for data analysis across different domains.

Introduction

- Increasing data volumes require efficient tools for analysis and visualization.
- Traditional workflows require multiple tools and technical expertise.
- Existing AI-assisted systems address parts of data analysis and visualization, but AgentIQ aims to unify the analytical workflow.
- AgentIQ combines multilingual natural-language query understanding,adaptive workflow selection, multi-agent analysis, and interactive dashboard generation.
- Specialized agents handle data understanding, planning, analysis, validation,, visualization, and explanation.

Motivation:

Build a unified, adaptive, transparent, and domain-agnostic natural-language data analytics framework.

Literature Review/ Background Study

Existing Research	Limitation / Gap	AgentIQ Contribution
Phebe (2026) – Multi-agent natural-language interface for data cleaning, Text-to-SQL, analysis and interactive visualization.	Mainly targets CSV/tabular data and plain-English interaction.	Supports multilingual queries and heterogeneous file/database sources with a unified analytical workflow.
MultiVis-Agent (2026) – Multi-agent framework for reliable multimodal visualization with logic-based coordination.	Focuses mainly on visualization reliability and multimodal refinement.	Extends the workflow with query understanding, analysis schema, planning, privacy and explainable analytics.
DeepEye (2026) – Self-driving data-agent system for heterogeneous data and complex analytical workflows.	Strong autonomous data-analysis capability, but broad in scope.	Focuses specifically on multilingual query understanding + explicit analysis-schema-driven planning + interactive dashboards.
Data-to-Dashboard (2025) – Multi-agent LLM pipeline for domain detection, insight generation and dashboard creation.	Mainly oriented toward domain-aware business analytics.	Provides a domain-agnostic workflow across different data sources with multilingual querying and privacy-aware processing.
NVAGENT (2025) – Collaborative agents for natural-language-to-visualization over single and multi-table data.	Concentrates on NL-to-visualization, especially multi-table scenarios.	Extends from visualization to complete data analysis, adaptive planning, explanation and dashboard alternatives.

Problem Statement

• Data analysis across heterogeneous sources is often complex and requires multiple tools and technical expertise.

• Existing AI analytics/visualization systems address individual parts of the workflow but struggle with complex analytical requirements and multilingual interaction.

• Limited transparency in how analytical workflows and visualizations are selected can reduce user trust.

• A unified framework is needed to combine multilingual query understanding, adaptive analysis, heterogeneous data handling, privacy, explainability, and interactive dashboards.

Problem to Solve:

Develop AgentIQ as an adaptive multi-agent framework for multilingual natural-language analytics and explainable interactive dashboard generation.

Project Scope

The scope of this project includes:

- Understanding user queries and problem statements.
- Accepting different types of datasets as input.
- Using multiple AI agents to perform different tasks.
- Analyzing the given data automatically.
- Selecting suitable charts and visualizations.
- Generating dashboards without manual work.
- Providing insights based on the analyzed data.
- Allowing users to interact with the system through a conversational interface.
- Supporting different domains such as education, healthcare, business, and research.

Data Gathered/ Data Used

Data Sources:

Source Type	Data Sources
Files	CSV, Excel, JSON
Cloud Spreadsheet	Google Sheets
Relational Databases	MySQL, PostgreSQL
NoSQL Database	MongoDB

Planned Evaluation Domains

- Education
- Healthcare
- Business
- Sales

Data Characteristics

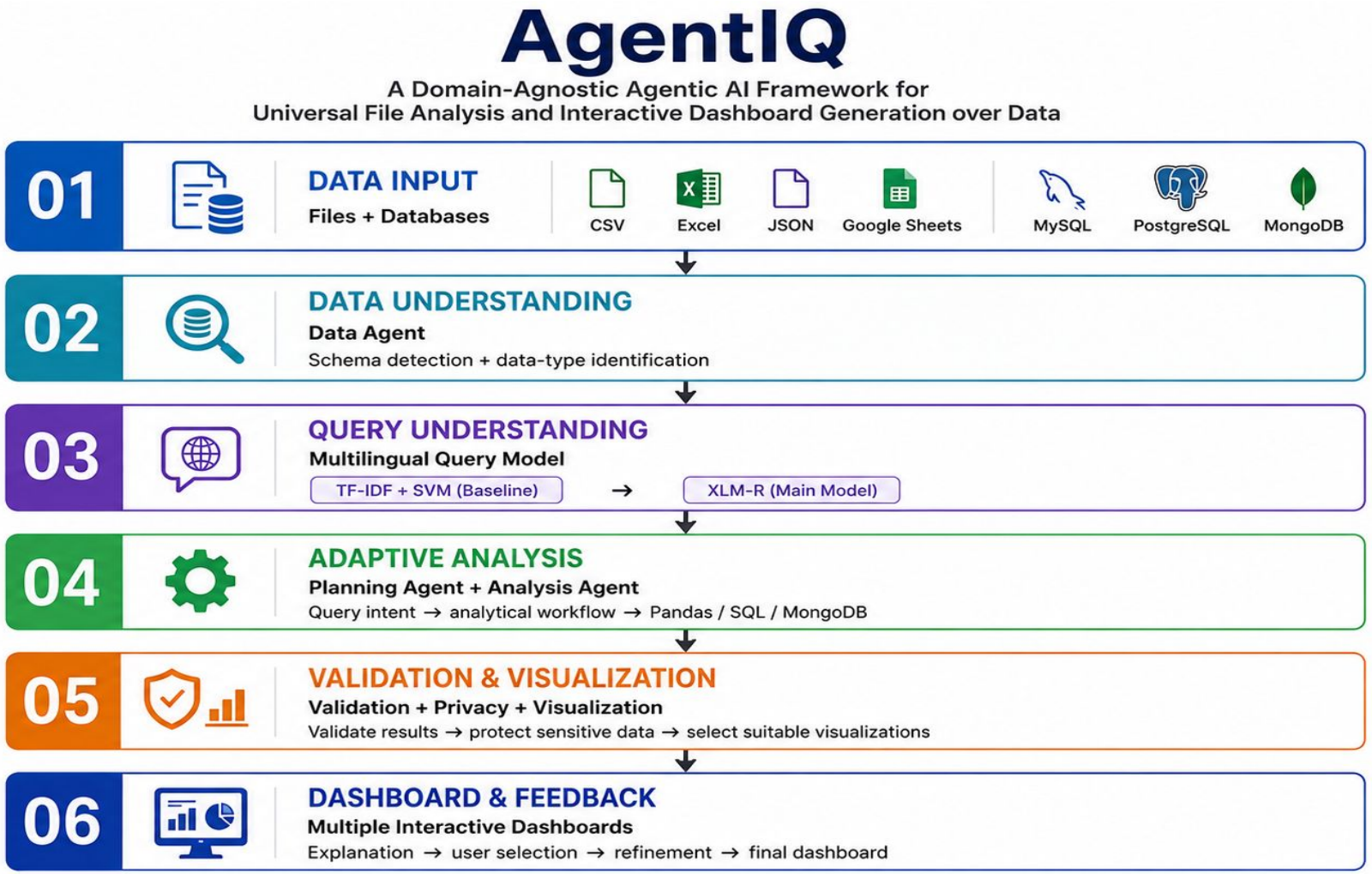
- Numerical attributes
- Categorical attributes
- Date/Time attributes
- Text attributes, where applicable
- Sensitive attributes, where applicable

Methodology / System Design

USER
↓
DATA SOURCES
↓
DATA AGENT
↓
QUERY UNDERSTANDING AGENT
↓
ANALYSIS SCHEMA
↓
PLANNING AGENT

ANALYSIS AGENT
↓
VALIDATION AND PRIVACY
↓
VISUALIZATION AGENT
↓
MULTIPLE DASHBOARDS
↓
EXPLANATION & REFINEMENT
↓
FINAL DASHBOARD

Methodology/ System Design(cont.)



Implementation / Modules

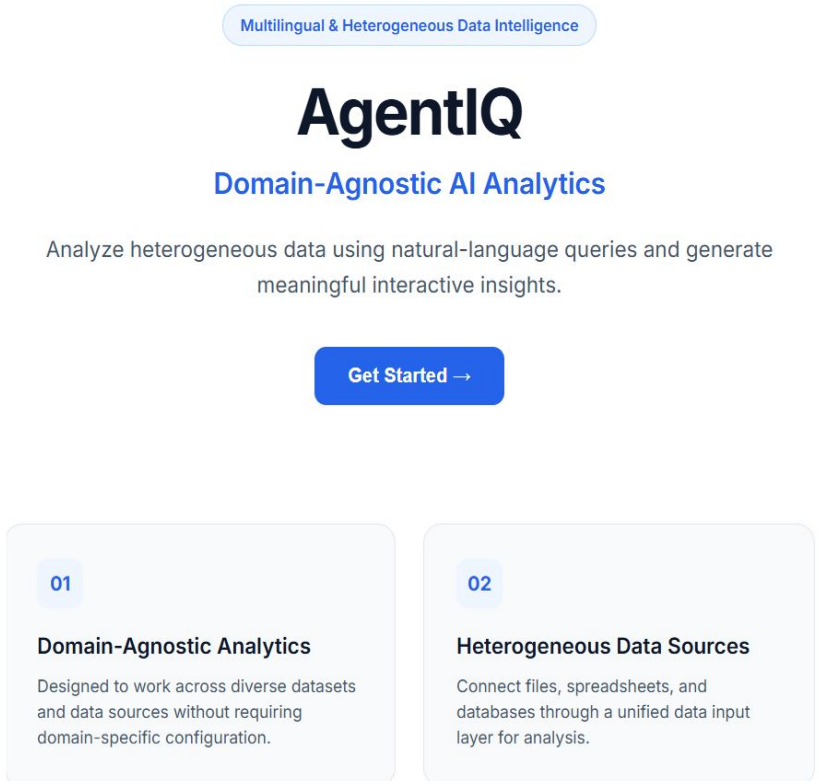
```
from fastapi import FastAPI, UploadFile, File, HTTPException
from app.data_input.service import DataInputService
from app.data_agent.agent import DataAgent
from app.auth.router import router as auth_router

app = FastAPI(title="AgentIQ API", version="1.0.0")
app.include_router(auth_router)

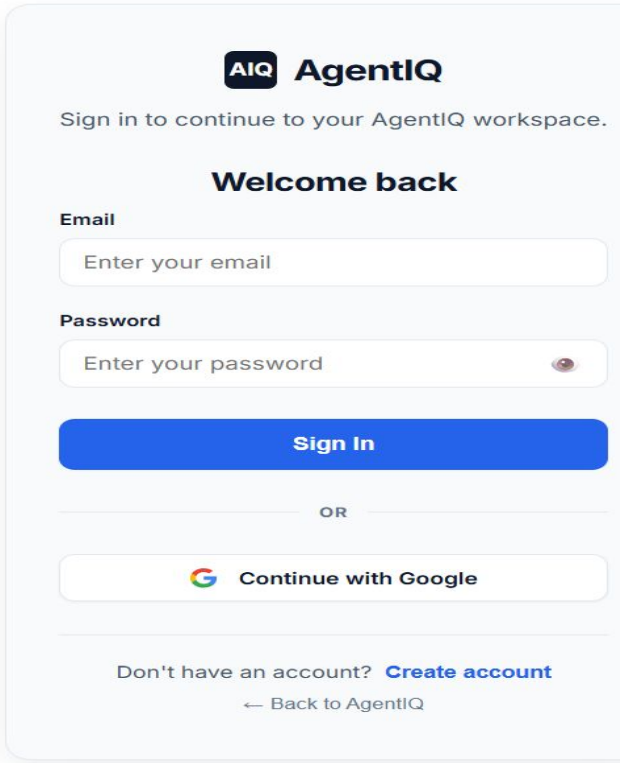
@app.get("/")
def root(): return {"message": "AgentIQ backend is running"}

@app.post("/api/v1/profile")
async def profile_file(file: UploadFile = File(...)):
    try:
        t = DataInputService.detect_file_type(file.filename)
        df = DataInputService.load_data(t, file.file)
        return DataAgent().profile(df)
    except Exception as e: raise HTTPException(400, str(e))
```

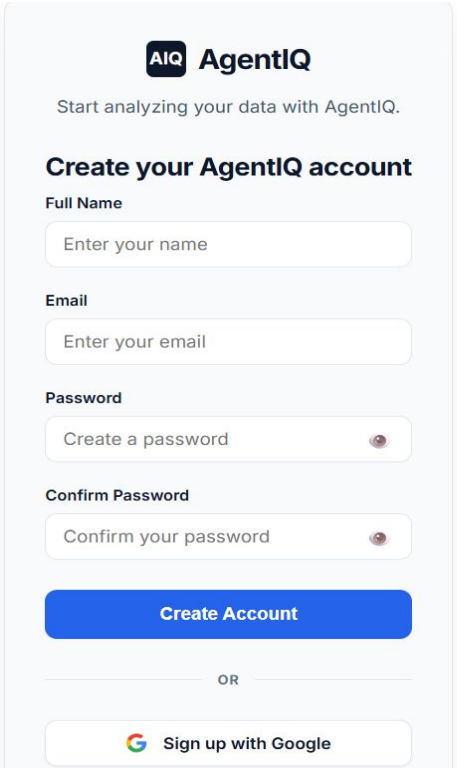
Results (Full) / Screenshots



LANDING PAGE

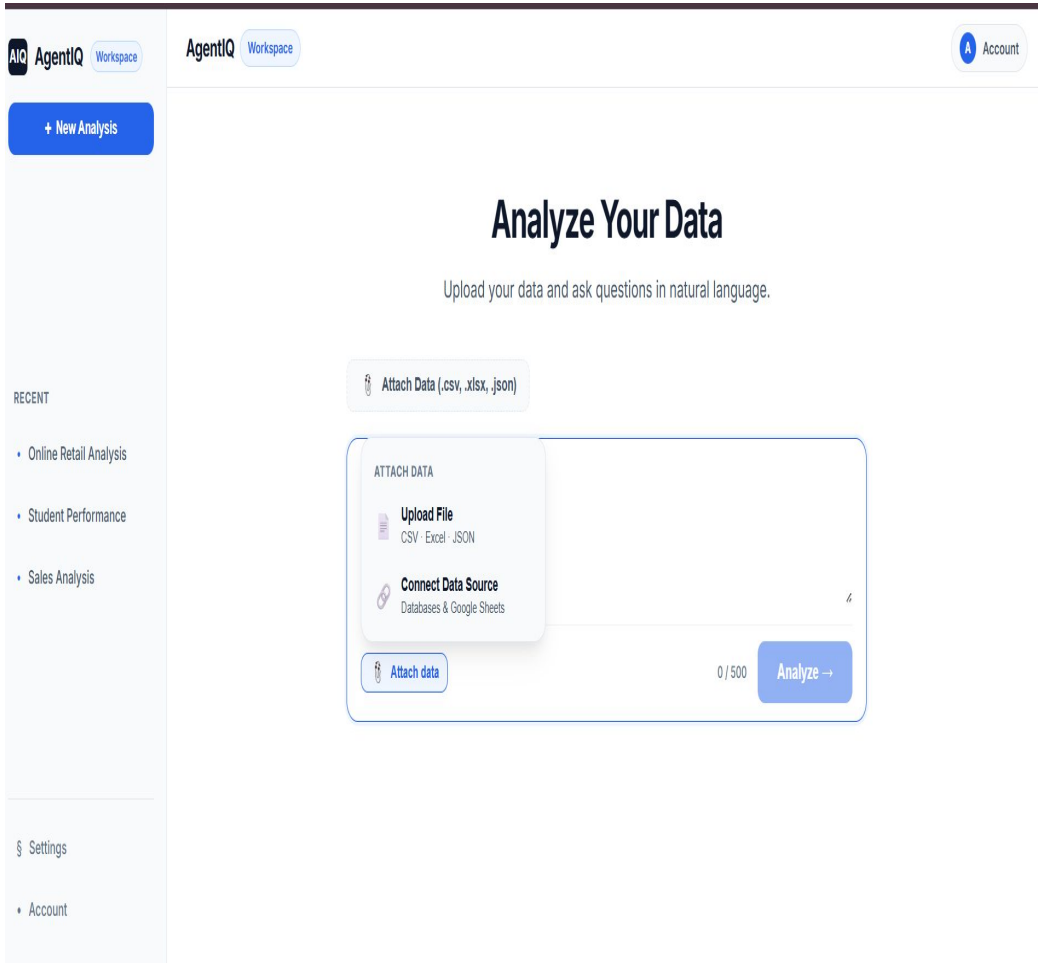


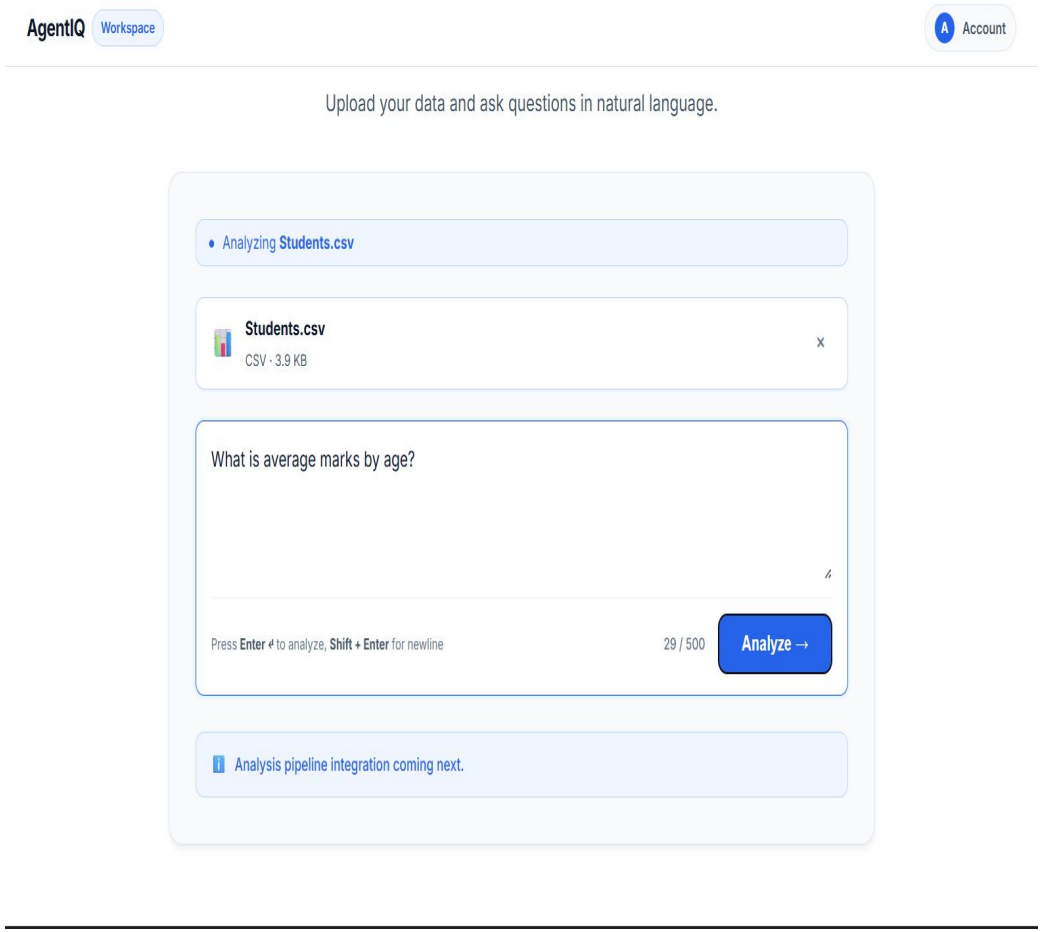
LOGIN PAGE



SIGN IN

Results (Full) / Screenshots





Impact Assessment

Technical Impact

- Automates data analysis and visualization.
- Reduces manual effort through multi-agent processing.
- Generates interactive and explainable dashboards.
- Supports multiple data sources and natural-language queries.

Social Impact

- Makes data analysis easier for non-technical users.
- Multilingual interaction improves accessibility.
- Useful across education, healthcare, business, and research.

Economic Impact

- Reduces time required for data analysis.
- Minimizes dependency on multiple analytical tools.
- Improves productivity through automated dashboard generation

Challenges Faced

- **Handling Heterogeneous Data**

  Different data sources may contain different structures, formats and data types. A common preprocessing and data-understanding workflow can be used to standardize the input.

- **Natural-Language Query Understanding**

  User queries can be ambiguous or expressed differently. Structured query interpretation can be used before sending tasks to analytical agents.

- **Multi-Agent Coordination**

  Different agents must exchange outputs correctly and execute tasks in the proper sequence. A controlled workflow can coordinate the agents.

- **Automatic Visualization Selection**

  Different datasets require different charts. Visualization selection rules can consider data types and analytical intent

Conclusion

AgentIQ proposes a domain-agnostic multi-agent framework for simplifying data analysis and interactive dashboard generation. The system is designed to accept heterogeneous data sources, understand natural-language and multilingual queries, coordinate specialized AI agents, perform analysis, select suitable visualizations and generate explainable dashboards. By integrating these stages into a unified workflow, AgentIQ aims to reduce the technical effort required for data analysis. The framework is designed for use across multiple domains such as education, healthcare, business and research.

Future Work

- Integrate real-time data streaming for continuously changing datasets.
- Add voice-based interaction along with text-based querying.
- Improve multilingual support for more languages and complex queries.
- Improve agent coordination and validation for complex analytical tasks.
- Enhance dashboard customization and recommendation capabilities.
- Improve scalability for larger datasets and enterprise applications.
- Strengthen privacy and security mechanisms for sensitive datasets

References

1. Phebe: A Multi-Agent Natural Language Interface for Interactive Data Visualization (2026)
   Paper: Phebe – ACM
   Relevant to our natural-language querying, multi-agent analysis, data cleaning, and interactive dashboard generation.
2. MultiVis-Agent: A Multi-Agent Framework with Logic Rules for Reliable and Comprehensive Cross-Modal Data Visualization (2026)
   Paper: MultiVis-Agent – Research Paper
   Relevant to our multi-agent visualization, validation, reliability, and iterative refinement.
3. DeepEye: A Steerable Self-driving Data Agent System (2026)
   Paper: DeepEye – arXiv
   Relevant to our heterogeneous data sources, autonomous analytical workflows, dashboards, and human-in-the-loop analysis.
4. Data-to-Dashboard: Multi-Agent LLM Framework for Insightful Visualization in Enterprise Analytics (2025)
   Paper: Data-to-Dashboard – Paper
   Relevant to our multi-agent data analysis, domain-aware reasoning, insight generation, and automated dashboards.
5. NVAGENT: Automated Data Visualization from Natural Language via Collaborative Agent Workflow (2025)
   Paper: NVAGENT – arXiv
   Relevant to our natural-language-to-visualization, multi-agent planning, heterogeneous data, and validation.

THANK YOU